Irina H. Fylyppova

Senior Frontend Developer

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GitHub | LinkedIn

Kharkiv, Ukraine | Remote only | UTC+2 (Kyiv), working across any timezone

Open to: Senior Frontend Engineer · Full-Stack Engineer (Vue / Node) · Frontend Architect · Developer Advocate · Technical Writer — contract or freelance

Project catalogue — every project at full length

Professional Summary

Senior Frontend Developer with 8+ years building high-performance web applications — and the architecture underneath them. I take frontend estates end to end: at DGtek I was the sole frontend engineer across 14 repositories and ~72,000 lines of Vue, three applications over 100k+ customer and building records, with the entire data layer behind a Web Worker boundary and monthly revenue, churn and ARPU reporting moved off the server into IndexedDB cursors in the browser. 39 published npm packages, several of them shipped to third-party companies — one team's component became another company's website. I also write and teach: an open JavaScript course in three languages, on a rendering platform I built from scratch and release through a build gated by eight command-line checks.

Technical Stack

Frontend:
Vue 2/3 (Composition API, `<script setup>`) • TypeScript • Vuetify 2/3 • Pinia • Vuex • Vue Router • Vue i18n • Vite • Vitest • Cypress

Performance & Data:
Web Workers • IndexedDB (with advanced indexing) • REST API • WebSockets • Real-time Synchronization • Geolocation APIs (Google Maps, MapLibre GL, Geoscape)

DevOps & Tools:
Git • GitHub Actions • GitLab CI • pnpm workspaces • Cloudflare Workers / D1 / KV • Cloudflare Pages • Firebase Auth • Firestore • Telegram Bot API • vite-ssg • Directus

Backend:
Node.js • Express • Cloudflare Workers (D1, KV, Cron Triggers) • SQL migrations • REST API design • Token verification and RBAC

Design & Testing:
Figma • SEO • Accessibility (a11y) • E2E Testing (Cypress, Playwright) • Unit Testing (Vitest) • Type checking (vue-tsc)

Professional Experience

Senior Frontend Developer — ITNET

Lviv, Ukraine | Feb 2025–Present · *Remote*

Eight projects on Vue 3 and TypeScript, from marketing sites to the platform below.

- **VD Express Platform** — a white-label booking core as a pnpm monorepo split by responsibility (`wizard-core`, `app-types`, `app-services`, `shared-ui`, `event-manager`). The core imports no host store: the application registers them through `registerStoresFor*()`, so one core serves several independent carriers. Step order is declarative too — `WizardStep` predicates rather than a hard-coded flow.

- **VD-Express & Site Constructor** — international coach booking: real-time seat availability over WebSocket with reconnection and ping monitoring, interactive cabin plans, door-to-door transfers on MapLibre, runtime-loaded themes for new brands without a rebuild, Firebase Auth, three locales.
- **BUSPORTAL** — a marketing site statically generated with `vite-ssg`: 20+ pre-rendered routes across four locales, Pinia state serialised into the markup, and the SEO layer extracted into a standalone composable.

Stack: Vue 3, TypeScript, Vuetify 3, Pinia, Vue Router, vue-i18n, Vite, Vitest, Cypress, MapLibre GL, Firebase Auth, WebSocket, pnpm workspaces

→ **All eight projects in the catalogue**

JavaScript & Vue.js Instructor — Ukrainian IT School

Kharkiv | Sep 2019–Sep 2025 · *Part-time, alongside client work*

Taught JavaScript and Vue.js to junior developers for six years, on a course I wrote and still maintain — and a shorter Vue course beside it.

- Fundamentals through to the awkward parts: functions as values, objects and constructors, references, prototypal inheritance, `this` and the ways it is lost.
- Vue 2 and Vue 3 with the Composition API, state management, REST APIs and project work.
- Students were onboarded onto commercial projects, which is the part that decides whether a course was any good.

The course outgrew the classroom and is now **Client-Side JS** — 164 lessons in three languages, described below.

→ garevna.github.io/js-lessons

Front-End Instructor — A-Level IT School

Kharkiv | 2017–2019 · *Part-time*

Where the course began. Two years of teaching front-end fundamentals, with the material written as it was taught — the first public release of it dates from 2017.

Senior Frontend Developer / Frontend Lead — DGtek Fibre Network

Australia | Sep 2019–Sep 2023 · *Remote*

Sole frontend engineer on a B2B provisioning platform for a fibre operator — 14 repositories, ~72,000 lines of Vue, three applications and 7 in-house npm packages, over 100k+ customer and building records. Four kinds of user: the operator's admins, the reseller ISPs selling on its network, the public checking an address, and the shareholders, who read the reports.

- **Architected the Web Worker data layer** shared by all three applications: a route/action message controller, credentials encrypted rather than left in localStorage, all API traffic, and a versioned IndexedDB cache of ten object stores with secondary indexes and migration-aware upgrades — which is what kept a heavy admin UI responsive and offline-tolerant.
- **Moved the reporting the shareholders read off the server**: monthly revenue, churn and ARPU are computed in a worker over IndexedDB cursors and drawn with Google Charts, removing a whole class of slow report endpoints.
- **Turned a component into a distribution channel**: coverage-check packages shipped to DGtek's reseller ISPs draw the operator's live polygons on sites DGtek does not control, resolve a visitor's address against them and quote a realistic connection time from that zone's settings. Coverage is edited in one place and every downstream site follows.
- **Packaged the address search**, which the worker boundary made possible: it geocodes against two sources — Geoscape for Australian addresses, Google Maps otherwise, with a fallback for addresses Google cannot resolve — then runs point-in-polygon on whatever coordinates come back to decide which coverage zone the building falls in. Self-contained behind a message port, so it shipped as a published package and ran on sites DGtek did not control.
- A fibre-haul cost calculator pricing a run to the nearest pit and exporting a client-ready PDF quote, and bulk import/export tooling with a validation pipeline.

Stack: Vue 2, Vuetify 2, Vuex, Web Workers, IndexedDB, Service Workers/PWA, Webpack, Google Maps API, Geoscape API, crypto-js, npm publishing

→ **Full case study** · **Catalogue**

Frontend Developer — Pineapple

ISP reselling DGtek's network | 2019–2022 · *Remote*

A Nuxt 2 public site, statically generated and composed from **10 npm packages I published**, with selective hydration through `vue-lazy-hydration` and navigation resolved from the CMS at runtime — plus the headless CMS itself (Vue frontend, Node/Express backend) storing landing pages as JSON with a publish/hide lifecycle.

Stack: Nuxt 2, Vue 2, Vuetify 2, Vuex, vue-lazy-hydration, PWA, Node.js, Express

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Independent Products & Education

Full-Stack Developer — Bible

Reading app and ministry coordination platform | 2023–Present · kcc-bible.pp.ua · source

Started alone in 2023 as a Bible reader and grown into the tool a church runs its ministries on. Now developed with an AI coding agent — the practice above comes from here. About 25,000 lines; my only TypeScript-first codebase, and the one I would show to explain how I work today.

- **A serverless back end of its own**: a Cloudflare Worker verifies the caller's Firebase token and decides authorisation itself — roles and ministries in KV, messages, events, participants and materials in D1 behind five versioned SQL migrations, a nightly cron, a CORS allowlist, the bot token as a Worker secret. 3,600 lines of Worker TypeScript with 1,700 lines of tests beside it.
- **Permissions designed against lock-out**: reader, publisher, administrator, plus owners the admin page cannot modify, and no administrator can change their own role.
- **A rota with real operational logic**: recurring events with a different time per weekday and single-day overrides, two invitation modes, substitutes who agree in advance and step in automatically, escalation to the schedulers when both decline.
- **Two Telegram bots** — one publishing to the church channel, a service bot writing to ministers privately with inline buttons whose effect matches acting in the app.
- **A rewrite that kept its users**: a migration script mints a Firebase Auth account per legacy login, dry run by default, with `--check`, `--undo` and `--delete-legacy` as separate deliberate steps.

Stack: Vue 3, TypeScript, Pinia, Vuetify 3, Vite, PWA, Cloudflare Workers / D1 / KV / Cron, Firebase Auth, Firestore, Telegram Bot API, MapLibre GL, Playwright, Vitest, pnpm

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Author & Platform Engineer — Client-Side JS

Independent | 2017–Present · garevna.github.io/js-lessons

The course from the two schools above, rebuilt as a product: 164 lessons in Ukrainian, Russian and English, on a rendering platform I wrote myself. Public and deployed since 2017 — garevna.github.io, vue-course (2018), js-simulator (2018), js-course (2019) and Client-Side JS today.

- **A rendering platform with no framework**: native custom elements with shadow DOM, a versioned service worker for offline reading, and web workers delivering content and icons.
- **An authoring language and its compiler**: lessons compile to components — replayable console and terminal sessions, quizzes, and code-repair exercises that run the reader's code in a sandboxed frame and score the repair.
- **Release gated by eight command-line checks**, including a byte-exact translation round-trip: every page is reassembled from its skeleton and message files and compared with the original before anything ships.

- **A serverless rebuild under a deadline**: when Glitch ended project hosting, the five practice back-ends twelve lessons depend on were rebuilt as one Cloudflare Worker, every route reconstructed from the request text printed in the lessons.

Stack: JavaScript (ES2022), Web Components, Shadow DOM, Service Workers, Web Workers, Webpack, Cloudflare Workers, Node.js tooling, Chrome DevTools Protocol

→ **Catalogue**

Core Competencies

Performance & data at scale — Web Workers and IndexedDB with secondary indexes and migration-aware upgrades; reporting computed over cursors in the browser instead of on the server; bundle splitting and lazy hydration.

Architecture — white-label and multi-brand systems configured at runtime; monorepos split by responsibility with dependency inversion, so one core serves several hosts; plugin-based reuse; state split into subject-specific stores.

Real-time — WebSocket synchronisation with automatic reconnection, exponential backoff and ping monitoring; multi-app data sync; offline-first fallbacks.

Back end — Cloudflare Workers with D1, KV and cron triggers; versioned SQL migrations; token verification and role-based access designed against privilege lock-out; REST APIs on Node and Express.

Accessibility & SEO — semantic HTML, ARIA, screen-reader testing; static pre-rendering with hydration; canonical, hreflang, Open Graph and generated sitemaps; dynamic meta tags for search engines and AI crawlers.

AI-Assisted Engineering

- **Agent-driven development with a verification loop:** a substantial share of my work is produced with an AI coding agent and none of it is trusted on sight — changes are exercised in a real headless browser over the Chrome DevTools Protocol, measured before and after, and reverted byte for byte when the measurement contradicts the claim
- **Harness before feature:** every new construct ships with a validator, so a malformed input is named on the command line instead of rendering as silent emptiness — the role an eval suite plays in an LLM product
- **Sandboxed execution of untrusted code:** reader-submitted code runs in an isolated frame and is graded by behavioural assertions rather than text comparison, because a real problem has several correct answers
- **Scoring instead of pass/fail:** an exercise can report that the code works and could have been written better, which is usually the truthful answer

Mentoring & Communication

- 10+ years mentoring junior developers, with the courseware public since 2017
- Onboarding students to commercial projects
- Direct stakeholder communication with C-level executives
- Technical documentation and code reviews — including documentation written for the non-technical people who use what I build

Education

Kharkiv National University named after V.N. Karazin
Higher Education (Mathematician)

Portfolio

39 published npm packages, and 87 public repositories of which 63 are deployed as live sites — Vue and JavaScript in roughly equal measure, with at least one release every year since 2017.

- 📦 **npm:** https://www.npmjs.com/~garevna
- 📚 **JS Lessons:** https://garevna.github.io/js-lessons/
- 🧠 **JS Quiz:** https://garevna.github.io/js-quiz/
- 🎨 **JS Philosophy Memes:** https://github.com/garevna/cv-resume/tree/main/media/js-memes
- 🔗 **GitHub:** https://github.com/garevna
- 💼 **LinkedIn:** https://www.linkedin.com/in/garevna/

Recommendations

Languages

- UA Ukrainian (Native)
- RU Russian (Fluent)
- GB English (C1 — Advanced)

Work Preferences

- ✅ Remote only
- 💰 Rate: Depends on scope and engagement — happy to discuss
- 📍 Timezone: UTC+2 (Kyiv)
- 🔄 Open to long-term engagements, complex projects, and architectural challenges